THE INTERPLAY OF INTELLIGENCE AND EDUCATION IN BRITISH SOCIAL CLASS MOBILITY: INSIGHTS FROM MEDIATION ANALYSIS OF FOUR CATEGORICAL VARIABLES

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Summary

This study examines the roles of IQ and education in shaping social mobility within British society, with a particular focus on their pathways and interactions. It investigates whether these attainments, considered either jointly or independently, contribute to social mobility across three distinct cohorts (1946, 1958, and 1970). The analysis employs a multinomial logistic model to quantify the effects with conditional probabilities, treating IQ and education as categorical variables. The study focuses on associations rather than causal effects. The findings reveal that none of the indirect pathways—whether mediated by IQ or education—account for more than 50% of the total effect, with no significant differences observed across cohorts or genders. Notably, IQ alone exhibits the least explanatory power but shows a more pronounced impact when combined with education, particularly in cases of long hierarchical distance mobility. This research provides a more nuanced understanding of the factors influencing social mobility, highlighting the complex interplay between intelligence, education, and class dynamics.

Keywords: Categorical data analysis, Multinomial logistic model, Mediation analysis

1 Introduction

1.1 Background

Social mobility refers to the movement of individuals across different social strata over time. This movement can be upward or downward, signifying a shift from a less advantaged to a more advantaged position or vice versa. (Bukodi and Goldthorpe (2018)) Social mobility has become increasingly crucial in contemporary society, especially in the context of health research, social cohesion, equity, and overall well-being. Recent studies have focused on the patterns of association between class of origin and class of destination, examining intergenerational class mobility. A significant body of research has analyzed the role of education—often seen as a key predictor of upward social mobility—within the framework of the 'OED triangle'(Bukodi and Goldthorpe (2016)). This concept highlights the pivotal role of education in promoting greater social openness, measured through three distinct associations: origin-education (OE), education-destination (ED), and origin-destination (OD) linkages, and it can be quantified by log-linear models (Vallet (2004)). Education is widely regarded as the cornerstone of individual economic and social mobility. However, studies on intergenerational social mobility have shown that although education mediates part of the relationship between origin and destination classes, it can't fully account for it.

Long before these recent discussions, Burt (1961) hypothesized that in a highly organized society, discrepancies between the general intelligence of children and the occupational class into which they are born are likely to produce a substantial and relatively consistent degree of "basic mobility." This idea has been supported by further research, such as that conducted by the National Institutes of Health (NIH), which suggests that social, educational, cultural, and physical factors during childhood significantly influence an individual's movement between social classes throughout their life. Among these factors, IQ (intelligence) has emerged as a critical determinant in assessing whether it can serve as an 'engine of social mobility.'

Previous studies, such as that by Deary et al. (2005), have established a robust correla-

tion between IQ and educational achievement, demonstrating that higher childhood IQ is associated with greater educational attainment (also noted by Connelly and Gayle (2019); Vanhanen (2011); Anni and Mõttus (2019)). However, this relationship is also significantly mediated by familial social circumstances. Scholars like Hassler and Mora (2000) have debated whether IQ should be primarily viewed as a form of genetic inheritance or as a product of social heritage, recognizing that environmental factors, such as the number of books in the home, also play a crucial role in enabling individuals to realize their potential (Thienpont and Verleye (2004)). Further research has dissected IQ by examining its components, such as verbal and spatial performance, with findings indicating a positive correlation between these components and social status. Johnson et al. (2010) examined the environmental interactions on IQ ability and demonstrated that different subaspects of cognitive ability may account for varying social capabilities. Forrest et al. (2011) conducted a comprehensive study that combined childhood IQ and educational factors to investigate social mobility, finding that both childhood IQ and education level independently contribute to upward social mobility between the ages of 49-51 years. Notably, education alone was positively associated with upward mobility between the ages of 5 and 25 years, while IQ alone was more strongly linked to upward mobility between the ages of 25 and 49-51 years. More recently, Connelly and Gayle (2019) introduced the concept of the "Flynn effect" as an explanatory variable in IQ development. Building on Gayle's research, Paterson (2021) conducted a longitudinal analysis extending to children born from 1936 to 2002. Deary et al. (2005) also found that different studies using the same dataset may yield varying results, potentially due to different approaches to data handling. In summary, these findings underscore the importance of considering multiple factors, including IQ, when analyzing the mechanisms that link class of origin to class of destination.

1.2 Research Question

This study aims to discern the extent to which observed social mobility can be attributed to class-based disparities in educational and intellectual (IQ) attainments. Specifically, it examines whether these attainments, considered jointly or indepen-

dently, account for mobility within British society, as evidenced across three distinct cohorts (1946, 1958 and 1970 cohorts). The research seeks to elucidate the roles that education and intelligence (IQ)—as mediating factors between one's class of origin and class of destination—play in social mobility. Moreover, the study delves into methodological approaches to effectively define and quantify these mediators, particularly when they are categorized as discrete rather than continuous variables.

Previous research by Kuha et al. (2021), which examined education as a mediator, found that indirect effects account for, on average, less than half of the total effects between an individual's class of origin and their destination class. This suggests that educational attainment explains less than fifty percent of the observed intergenerational mobility. The influence of education on social mobility has remained relatively stable over time, with a potentially greater impact on women than on men. Additionally, the contribution of education to mobility varies significantly across different segments of the class structure. Building on Kuha's work, the research further explores this area by incorporating additional variables IQ to provide a more comprehensive understanding of the factors influencing social mobility.

Figure 1 illustrates the mediation framework employed in this study, which considers two mediators: education and intelligence (IQ). The model examines the direct and indirect pathways through which class of origin influences class of destination. Specifically, the paths labeled a to f, represent the various relationships between the variables. These effects can be divided into absolute and relative associations. Absolute associations are derived from the marginal distributions of class of origin and class of destination, while relative associations are calculated using log odds ratios, which measure the strength of association independent of the marginal distributions. These associations will be explicitly defined and further elaborated upon in Section 5.

Methodologically, the study begins with the construction of contingency tables that cross-classify individuals by their class of origin and destination. Employing linear regression to analyze continuous variables offers a straightforward approach, relying on

model coefficients for estimation. However, this study opts for a more comprehensive method to evaluate categorical data. When dealing with categorical variables, the analysis utilizes conditional distributions based on fixed values in a population, such as the distribution of the outcome variable (Y) conditioned on the exposure variable (X). For indirect associations, the focus shifts to the distribution of the mediator (M) dependent on X, aligning with the reference distribution of X itself. In certain scenarios, if the mediator M is continuous, summations are replaced by integrals. Similarly, if the outcome Y is continuous, probabilities $P(Y|X, M)$ are substituted with expectations $E(Y|X, M)$. This paper will focus on categorical variables analysis, and joint mediators will not be explored; instead, education and IQ will be treated as distinct variables, allowing us to apply methods similar to those used in previous research involving three variables.

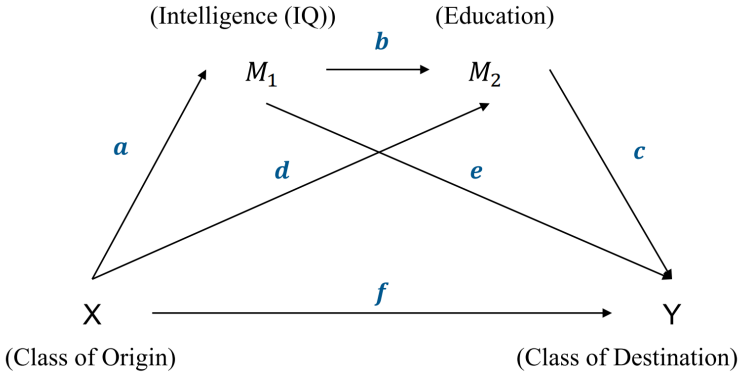


Figure 1: Mediation with Two Mediators

The dissertation is organized as follows: After the introduction, Section 2 delves into the data and variables, outlining their significance and context. Section 3 is dedicated to the development of the model and the derivation of the key research formulas. Section 4 presents a simulation study designed to validate the model and ensure the accuracy of the codes. Section 5 is the core empirical study, which involves analyzing different cohorts and genders, calculating the confidence intervals for the estimated ratios using bootstrap methods, and applying a regression model to identify factors influencing these ratios. Finally, Section 6 provides a summary of the research findings, along with the study's limitations and suggestions for future research improvements.

2 Data and Variables

The data is from three British birth cohort studies conducted in 1946, 1958 and 1970 respectively. A birth cohort consists of individuals born within a specific time frame who share common characteristics. For instance, the 1946 cohorts are composed of individuals born in the same week of March 1946, akin to similar cohorts established in the UK in 1958 and 1970. Specifically, the 1946 cohort is known as the National Survey of Health and Development (NSHD), the 1958 as the National Child Development Study, and the 1970 as the British Cohort Study. These cohort studies are of paramount importance as they chronicle the lives of participants from birth through childhood, adolescence, and adulthood, into their retirement years. By tracing these life trajectories, the research can identify the factors that contribute to variations in individual quality of life and discern the interrelationships between diverse influences.

The dataset includes four key variables: class of origin (X), IQ (M_1), education (M_2), and class of destination (Y). Social class is primarily categorized into seven groups based on the National Statistics Socio-Economic Classification (NS-SEC), the main framework used in British official statistics.

Class destination (Y) and class origin (X). The term 'Destination' refers to the social class position of the respondent at the age of 38 years, or the last known employment status prior to this age. 'Origin' delineates the class position of the respondent's father when the respondent was 10 or 11 years old, or at 15 or 16 years old if data from the earlier age is unavailable. The classification encompasses: Higher managers and professionals (1), Lower managers and professionals (2), Intermediate occupations (3), Small employers and own account workers (4), Lower supervisory and technical occupations (5), Semi-routine occupations (6), and routine occupations (7). The number of each classifictions will be used in section 5 for empirical results. It is imperative to note that hierarchical comparisons, such as assuming higher managers possess more advantages than small employers, are inadvisable. The data should be treated as unordered categorical data. Consequently, this classification facilitates the construction of a 7×7 contingency table representing various combinations of class

origin and class of destination.

Variable education (M_2). The educational qualifications is provided by the eight ordered categories that ranging from "no qualifications" to "postgraduate qualifications". However, we collapse the eight categories of educational qualifications into four to produce relative levels. And the number of the original categories divided in different way in different cohort, and the groups approximately match the quartiles of qualification levels within the cohort(Bukodi and Goldthorpe (2016)). In short, one could say that as the proportion of cohort members with degrees increases, other qualifications tend to be pushed down the relative levels.

Variable IQ (M_1). An important aspect of this research is the integration of the IQ variable into the social mobility data, ensuring proper alignment by individual identity. In the National Survey of Health and Development (NSHD) cohort (1946 cohort), intelligence assessment began at the age of eight, as noted by Richards et al. (2019). The assessments included verbal and nonverbal ability tests developed by the National Foundation for Educational Research, covering (1) reading comprehension, (2) word reading, (3) vocabulary, and (4) picture intelligence tests. The individual scores from these tests were standardized to a mean of 0 and a standard deviation of 1, combined to create a composite intelligence score, and then re-standardized. Although the specific IQ testing standards for the 1958 and 1970 cohorts were not found, the characteristics of the data suggest that they are likely similar to those of the 1946 cohort.

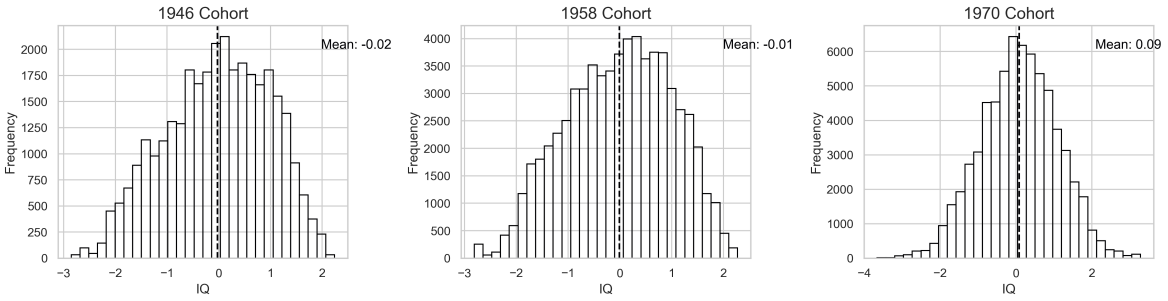


Figure 2: IQ Distributions in Different Cohorts

The variable has already been standardized. From the Figure 2, it is shown that with

the year grows, the average of IQ increases from -0.02 to 0.09. In 1946 cohort, the range of IQ scores from a low of -2.853 to a high of 2.238 shows a wide range of variability, covering very low to very high levels of intelligence. The median of 0.043 is almost equal to the mean, suggesting that the distribution of the data may be symmetrical and not significantly skewed. The distribution of IQ scores for 1958 and 1970 cohorts appear to be balanced and symmetrical, with no apparent skewness as well.

Therefore, we assume it approximates a normal distribution, and we divide the IQ data into five equal parts, labeled class1 through class5. The bottom 0%-20% is designated as class 1, $20\% - 40\%$ as class 2, and so forth. Table 1 reveals that the distribution of IQ classes among women remains similar between the years 1946 and 1958. In contrast, the distribution for men is concentrated more in the first and second classes, indicating a generally lower level of cognitive development compared to women. However, a shift is observed in the 1970 cohort where there is an enhancement in men's cognitive development, with a greater distribution in the fourth and fifth classes, while women's IQ distribution tends to shift towards lower classes.

Table 1: Marginal Distribution of Class by Cohort and Gender

Class	**1946 cohort**		**1958 cohort**		**1970 cohort**	
	Men	**Women**	**Men**	**Women**	**Men**	**Women**
1	23.58%	18.14%	23.09%	19.53%	16.58%	18.64%
2	19.43%	19.26%	20.56%	17.89%	19.89%	22.19%
3	19.83%	19.37%	18.62%	19.32%	21.35%	21.13%
4	19.04%	19.71%	18.89%	20.43%	20.86%	21.35%
5	18.11%	23.52%	18.34%	22.82%	21.32%	16.69%
Total	100%	100%	100%	100%	100%	100%

The missing data for the IQ variable amounted to 33.57% of the dataset. Specifically, the 1946 cohort had 14.47% missing values, the 1958 cohort had 44.62%, and the 1970 cohort had 27.21%. However, for this analysis, we will not use multiple imputation and will instead only use complete datasets for each model.(see in section 5).

3 Methodology

The methodology section begins with the introduction of a simpler three-variable model. This includes a discussion of the notation used, a review of relevant past research developments, and the formulation of the effect, which serves as the foundation for the more complex four-variable model. Following this, a detailed examination of the four-variable model is presented, elaborating on its structure and the methodology employed for its analysis.

3.1 Model Development

Three Variables Model. Firstly, consider the following variables in the process: the exposure X, the mediator M, and the outcome Y. Here, M can represent multiple mediators of interest, denoted as $M = M^{(1)}, M^{(2)}, \ldots, M^{(n)}$. To begin, let's examine the notation for the simplest case involving these three variables. $M_i(x)$ denotes the value of M when X is set to x for unit i, while $Y_i(x, m)$ represents the value that Y would take if the pair (X, M) is set to (x, m) for unit i.

The concept of correlation between two continuous variables can be traced back to Niles (1922), who introduced "path coefficients" as a mathematical criterion for assessing causal relationships between variables. Galton (1889) further developed the idea of mediators, identifying them as common causes that influence the variations in two traits, positioning them within the causal pathways that link independent variables to dependent variables. A commonly used method to estimate path coefficients involves employing a regression model with continuous variables, as described by MacKinnon (2000). This approach demonstrates that the indirect effect is equivalent to the total effect minus the direct effect. However, traditional methods have limitations, particularly in that both the independent and dependent variables must be continuous (as noted by Agresti (2012)), and they face challenges when applied to models with multiple mediators. Mediation analysis for continuous response variables is well established, and Samawi et al. (2018) extended these methods to categorical responses, particularly focusing on cases where both the dependent variable and the mediator

are dichotomous. In her approach, logistic regression models are employed to estimate the coefficients. The significance of the model is evaluated by assessing whether the difference in coefficients exceeds 20%. The corresponding logistic regression equations are expressed as follows: $logit(\pi_y(x)) = \alpha_1 + cx_i + e_{1i}$, $logit(\pi_M(x)) = \alpha_2 + ax_i + e_{2i}$, and $logit(\pi_M(x, m)) = \alpha_2 + c'x_i + bm_i + e_{3i}$.

The foundational work of Robins and Greenland (1992), subsequently built upon by Pearl (2022), sparked a proliferation of contributions that have since integrated formal language and estimation methods from causal inference, thus broadening the scope of such analyses under more explicit assumptions. Imai et al. (2010) proposed more versatile models (generalized linear models and generalized additive models), suitable for both outcomes and mediators. Yu et al. (2019) and Vemuri (2015) introduced general definitions of mediation effects applicable across all variable types (categorical or continuous) -for response, exposure, or mediation variables, presenting a transitional method for measuring causal effects at the unit level, often using $Y_i(r) - Y_i(s)$. Daniel et al. (2015) as well as Lange et al. (2014) proposed a more general method for mediation analysis, offering a concise overview of causal mediation estimators for a single mediator. This approach dissects the total causal effect (TCE) into the total natural indirect effect ($TNIE$) and the pure natural direct effect ($PNDE$), all defined by expectations. This theoretical framework has been extended by Steen et al. (2017), Taguri et al. (2018), and Vansteelandt and Daniel (2017). For instance, Daniel characterized the TCE between states r and s as $TCE = E(Y(r) - Y(s))$, $TNIE = E(Y_i(x^*, M_i(r)) - Y_i(x^*, M_i(s)))$, where the independent variable is fixed, and $PNDE = E(Y_i(r, M^*) - Y_i(s, M^*))$. However, this counterfactual framework—which allows for the decomposition of a total effect into a direct effect and an indirect effect, even when there are interactions and non-linearities—fits best when the exposure variable is binary, although it can also accommodate categorical data.

The literature offers more flexible methodology for causal mediation analysis, but the emphasis here shifts to scenarios where the parameters of interest are associative effects rather than strictly causal effects. The term "causation" is often used to describe

associative conditions, especially in the context of natural phenomena. The objective is to define and estimate population characteristics that yield informative descriptions of that population (Niles (1922)). Definitions of associations are aligned with those of causal mediation effects, with the key difference being that potential outcome distributions are replaced with conditional distributions of fixed variable values within populations.

Kuha and Goldthorpe (2010) conceptualized the indirect associations between X and Y using multinomial logistic models for cases where the categories exceed binary. The total effect is defined as the expected value of Y given that X equals r, expressed as:

$$E\left(Y|X=r\right)=E_M\left(E\left(Y|X=r,M\right)|X=r\right)=\int E(Y|X=r,M)P(M|X=r)dM \tag{1}$$

where the integral simplifies to a summation if M is categorical. The indirect effect can be defined as:

$$E^{IE}(Y|X=r)=\int \frac{E(Y|X=r,M)+E(Y|X=s,M)}{2}P(M|X=r)dM \tag{2}$$

To measure the unit difference, either the log odds ratio or the mean difference can be used. This approach allows for the decomposition of the total effect into indirect and direct components, thereby providing a deeper understanding of the associations within the population.

Kuha et al. (2021) further generalized equations (1) and (2), continuing to use the concept of the reference distribution, and derived a formula for a model with three variables in his recent work. For instance, he defines the indirect effect as

$$\pi_y{}^{IE}(x)=\sum_{x^*}\sum_{m}P(Y=y\mid X=x^*,M=m)P(M=m\mid X=x)P_0(X=x^*) \tag{3}$$

This formula can be straightforwardly divided into two steps. First, the conditional distribution $P(y\mid x)$ is given by $P(y\mid x)=\sum_m P(Y=y\mid X=x^*,M=m)P(M=m)$ where the mediator M is influenced by the exposure X. Second, the marginal distribution of M can be expressed as $P(M=m)=\sum_{x^*}P(M=m\mid X=x^*)P_0(X=x^*)$. By substituting the second equation into the first, the desired result is obtained. The

log odds ratio is used to assess the magnitude of influence. For instance, the value ${Q_{rs,tu}}^{IE} = \log\left(\frac{{\pi_u}^{IE}(s)/{\pi_t}^{IE}(s)}{{\pi_u}^{IE}(r)/{\pi_t}^{IE}(r)}\right)$ is calculated, where a larger value indicates a greater likelihood of transitions from s to u or r to t. This foundational work serves as the basis for deriving the formula for the model with four variables.

Four Variables Model. The discussion now shifts to the case where there are two mediators, which is the primary focus of this dissertation. This scenario involves the exposure X, the first mediator M_1, the second mediator M_2, and the outcome Y. The notation for the variables can be represented as follows: $M_1(x)$ denotes the value that the first mediator M_1 would take if the exposure $X = x$, with $M_2(x)$ being defined similarly. $M_2(x, m_1)$ represents the value that the second mediator M_2 would take if the exposure $X = x$ and $M_1 = m_1$, with $M_1(x, m_2)$ defined symmetrically. $Y(x)$ represents the value that the outcome Y would take if the exposure $X = x$, while $Y(x, m_1, m_2)$ denotes the value of Y when $X = x$, $M_1 = m_1$, and $M_2 = m_2$. Additionally, $Y(x, M_1(x^*), M_2(x^{**}, M_1(x^{***})))$ describes the value of Y when $X = x$, the first mediator is $M_1 = M_1(x^*)$, and the second mediator is $M_2 = M_2(x^{**}, M_1(x^{***}))$. The expression $(Y(x, M_2(x^*), M_1(x^{**}, M_2(x^{***}))))$ is defined asymmetrically. The principal contribution of this paper is the derivation of a formula for a four-variable model, extending the logic of the three-variable model.

The existing literature on multiple mediators from X to Y includes the following perspectives: (1) the multiple mediators do not affect one another, meaning the pathway from M_1 to M_2 is absent (as discussed by Preacher and Hayes (2008)); (2) the mediation involves a mediator-outcome confounder L, which is itself influenced by the exposure (VanderWeele and Vansteelandt (2014); VanderWeele and Chiba (2014), who also presented a sensitivity analysis). Imai and Yamamoto (2013) introduced the concept of multiple and related mediators, where M_1 can have effects on M_2 (also like Steen et al. (2017)). This situation will be also utilized in this study.

The simplest statistical model used is linear regression. MacKinnon (2000) characterized the total effect as $Y = \tau X + \varepsilon_1$. Additionally, the estimation of the effect

of the independent variable on each mediator is given by $X_{M_1} = \alpha_1 X + \varepsilon_2$ and $X_{M_2} = \alpha_2 X + \alpha_3 M_1 + \varepsilon_3$, where α_1 and α_2 represent the effects on each mediator, respectively, with α_3 representing the sequential effect. The effect from the mediators to the outcome is expressed as $Y = \tau' X + \beta_1 X_{M_1} + \beta_2 X_{M_2} + \varepsilon_4$.

Daniel et al. (2015) introduced a general approach for analyzing two causally ordered mediators, where M_1 can affect M_2, but not vice versa. This approach decomposes the total causal effect (TCE) between states r and s into four components: *NDE-sss*, NIE_1*-rss*, NIE_2*-rrs*, NIE_{12}*-rrr*. *NDE-sss* is defined as the natural direct effect not mediated by either M_1 or M_2, and is expressed as,

$$E\left\{Y(r, M_1(s), M_2(s, M_1(s))) - Y(s, M_1(s), M_2(s, M_1(s)))\right\}$$

NIE_1*-rss* is defined as the natural indirect effect through M_1 alone, which is given by

$$E\left\{Y\left(r, M_1(r), M_2\left(s, M_1(s)\right)\right) - Y\left(r, M_1(s), M_2\left(s, M_1(s)\right)\right)\right\}$$

NIE_2*-rrs* and NIE_{12}*-rrr* can be defined similarly,

$$E\{Y(r, M_1(r), M_2(r, M_1(s))) - Y(r, M_1(r), M_2(s, M_1(s)))\}$$

$$E\left\{Y(r, M_1(r), M_2(r, M_1(r))) - Y(r, M_1(r), M_2(r, M_1(s)))\right\}$$

By decomposing the total effect as described, Daniel demonstrates that the TCE can be expressed as the sum of these components: $TCE =$ *NDE-sss*$+$ NIE_1*-rss*$+$ NIE_2*-rrs*$+$ NIE_{12}*-rrr*. Vansteelandt and Daniel (2017) also presented a counterfactual framework to define the path. The method they used was through conditional expectations. For example, they defined the indirect effect via M_1 and M_2 as $E(Y_{rM_{1r}M_{2r}} - Y_{sM_{1s}M_{2s}}) = \sum_{m_1}\sum_{m_2}\left\{\left[E(Y|r, m_1, m_2) - E(Y|s, m_1, m_2)\right] P(m_1|s)P(m_2|s, m_1)\right\}$. This formula is derived from the *Power Property* of conditional expectation, which states that $E(X) = \sum_i E(X|A_i)P(A_i)$. However, another method, as proposed by Kuha et al. (2021), describes the indirect effects through summation over joint probabilities. This approach simplifies the description by focusing directly on the distributions, and will be the method used in the following analysis.

One more thing is needed to notice here, considering the different patterns of the two mediators M_1, M_2 case, the first pattern we can define is the sequential mediator $X \to M_1 \to M_2 \to Y$ (as in Vansteelandt and Daniel (2017)). However, in this case, it is possible to simplify it to a three-player condition. This is because there are only two distinct paths to analyze: the direct effect and the indirect effect. The mediators M_1 and M_2 can be viewed as a cohesive unit, enabling us to streamline our research by building upon previous studies. Additionally, while the relationship between the two mediators is considered, the focus remains on a unidirectional effect from M_1 to M_2. This approach avoids the complexities associated with analyzing bidirectional effects using regression models, as highlighted by Daniel et al. (2015). Therefore, the analysis in this context assumes that M_1 affects M_2, but not vice versa.

3.2 Definitions of Effects and Associations

3.2.1 Total Effect and Associations

For a unit i, the effect on Y of X being set to $X = s$ rather than $X = r$ is defined as a comparison between $Y_i(r, M_{1i}(r), M_{2i}(r, M_{1i}(r)))$, and $Y_i(s, M_{1i}(s), M_{2i}(s,\ M_{1i}(s)))$, which contains the path $a+b+c+d+e+f$ (all the path) in Figure 1. The analysis in this study focuses on estimating the aggregates within a population of units. Consequently, the distributions of $P(Y_i(r))$ and $P(Y_i(s))$ (equivalent to the longer expression discussed above) are examined. Equation (4) is used as a parameter to quantify the total effect:

$$\pi_y{}^*(x) = P(Y_i(x) = y) \tag{4}$$

The difference between r and s is used to calculate the effect,

$$\theta^{TE} = log\frac{\pi_u(s)/\pi_t(s)}{\pi_u(r)/\pi_t(r)} \tag{5}$$

3.2.2 Indirect Effect

In this paper, the indirect path can be classified into three distinct categories: $X \to M_1 \to Y$ (path $a + e$ in Figure 1), $X \to M_2 \to Y$ (path $c + d$), $X \to M_1 \to M_2 \to Y$ (path $a + b + c$), respectively. The following equation 6 - equation 8 will be defined to measure each indirect effects.

1. $X \rightarrow M_1 \rightarrow Y$

Definitions of indirect effects and direct effects just break the link between X and M_1 and only vary one of them. For example, a unit level indirect effect is to set $X = s$ and $X = r$, and to calculate the comparison of $Y_i(x^*, M_{1i}(r), {M_2}^*)$and $Y_i(x^*, M_{1i}(s), {M_2}^*)$. This demonstrates that changes in Y occur only in response to changes in M_1 when X changes, while X itself remains fixed. It is important to note that although the path only involves the mediator M_1, M_2 must also be held constant, resulting in a slightly different formula compared to the three-variable situation (equation (3)).

The following formula is used to quantify the indirect effect,

$$\pi_y{}^{IE}(x) = \sum_{x^*}\sum_{m_2}\sum_{m_1} P(Y = y|X = x^*, M_1 = m_1,\ M_2 = m_2)$$
$$P_0(M_2 = m_2)P(M_1 = m_1|X = x)P_0(X = x^*) \qquad (6)$$

For the sake of clarity in the formula, M_2 is positioned in the middle, where P_0 is the distribution of hypotheses that do not depend on other variables.

2. $X \rightarrow M_2 \rightarrow Y$

Applying the same logical framework as described above, equation (7) shifts the focus from mediator M_1 to mediator M_2. It highlights the difference between between the $Y_i(x^*, {M_1}^*,\ M_{2i}(r, {M_1}^*))$ and $Y_i(x^*, {M_1}^*, M_{2i}(s, {M_1}^*))$. Consequently, the following formula can capture the relationship:

$$\pi_y{}^{IE}(x) = \sum_{x^*}\sum_{m_1}\sum_{m_2} P(Y = y|X = x^*, M_2 = m_2, M_1 = m_1)$$
$$P(M_2 = m_2|X = x, M_1 = m_1)P_0(M_1 = m_1)P_0(X = x^*) \qquad (7)$$

3. $X \rightarrow M_1 \rightarrow M_2 \rightarrow Y$

In the scenario with a single mediator, it is possible to fix the value of X and examine how the varying effects of M influence the outcome Y. However, with the introduction of two mediators, this process can be viewed sequentially, where the influence on M_1 cascades to M_2, ultimately impacting Y. This approach allows for the construction of

the conditional probability of the model.

For example, a unit level of indirect ordered causal effect between $X = s$ and $X = r$ will be the comparison of $Y_i(x^*, M_1^*, M_{2i}(x^*, M_{1i}(r)))$ and $Y_i(x^*, M_1^*, M_{2i}(x^*, M_{1i}(s)))$, fixing the value $X = x^*$. In this situation, attention must be directed solely toward the inclusion of paths $a + b + c$, excluding path e) (as shown in Figure 1). This is because the indirect effect on Y is driven by changes in M in response to variations in X, while Xitself remains unchanged. The indirect effect is quantified using equation (8):

$$\pi_y{}^{IE}(x) = \sum_{x^*} \sum_{m_2} P(Y = y|X = x^*, M_1 = m_1{}^*, M_2 = m_2) \\ P(M_2 = m_2|X = x) P_0(M_1 = m_1{}^*) P_0(X = x^*) \tag{8}$$

Where $P(M_2 = m_2|X = x) = \sum_{m_1} P(M_2 = m_2|M_1 = m_1, X = x) P(M_1 = m_1|X = x)$

Specifically, $P(Y = y|X = x^*, M_1 = m_1^*, M_2 = m_2)$ captures the effect of M_2 on Y, conditional on X and M_1. Additionally, $P(M_2 = m_2|X = x)$ is decomposed into $P(M_2 = m_2|M_1 = m_1, X = x)$ and $P(M_1 = m_1|X = x)$, representing the effects of X on M_1 and M_1 on M_2, respectively. By summing these conditional probabilities, the total indirect effect across the pathway $a + b + c$ is obtained. Equation (9) represents the log odds ratio, which is calculated to capture the difference between r and s across the three categories mentioned above.

$$\theta^{IE} = log \frac{\pi_u{}^{IE}(s)/\pi_t{}^{IE}(s)}{\pi_u{}^{IE}(r)/\pi_t{}^{IE}(r)} \tag{9}$$

3.2.3 Direct Effect

Based on the results of the indirect effect, the scenario is mirrored by changing the value of X while keeping M_1 and M_2 fixed as the two mediators. This involves comparing the distributions Y across two groups, where the difference in Y is caused by the unit-level change between $X = s$ and $X = r$. Specifically, this comparison evaluates the difference between $Y_i(r, M_1{}^*, M_2{}^*)$ and $Y_i(s, M_1{}^*, M_2{}^*)$, with the same distribution $P_0(M_1 = m_1)$ and $P_0(M_2 = m_2)$. The following formula is then used to quantify the

effect:

$$\pi_y{}^{DE}(x) = \sum_{x^*}\sum_{m_2}\sum_{m_1} P(Y = y|X = x^*, M_2 = m_2, M_1 = m_1)$$
$$P_0(M_2 = m_2)P_0(M_1 = m_1)P_{x^*}(X = x^*) \tag{10}$$

Where $P_{x^*}(X = x) = 1$, the log ratio index is as follow,

$$\theta^{DE} = log\frac{\pi_u{}^{DE}(s)/\pi_t{}^{DE}(s)}{\pi_u{}^{DE}(r)/\pi_t{}^{DE}(r)} \tag{11}$$

In this mediation analysis, a key consideration is the decomposition of effects. The most straightforward approach involves partitioning the total effect into the sum of direct and indirect effects. However, this method is not suitable in this context because the interpretations of these effects are not inherently aligned. As a result, the three distinct indirect effects cannot simply be added together, nor can the direct effect be derived by subtracting the indirect effects from the total effect. Instead, the focus of the empirical study is on analyzing the ratios of the three distinct indirect associations to the total associations, rather than primarily examining a well-defined direct effect. This approach is consistent with previous research (Kuha et al. (2021)).

3.3 Estimation for Associations

When analyzing direct and indirect effects where both mediators (M) and outcomes (Y) are categorical variables, multinomial logistic regression provides a robust framework. This statistical method extends the binary logistic regression model to accommodate outcome variables with more than two categories, accommodating both continuous and categorical predictors. Considering now a response variable Y which has $L \geq 2$ possible categories ($L = 2$ is binary model), $l = 1, ..., L$. First, it is necessary to select the baseline level for the outcome of interest. For example, consider the probabilities $P(Y_i = l) = \pi_{il}$ for the multinomial distribution of a polytomous response variable Y_i for unit i. These probabilities can be specified as follows:

$$\log\left(\frac{\pi_{il}}{\pi_{i1}}\right) = X_i{}'\beta_j, \quad \text{for each } l = 2, \ldots, L. \tag{12}$$

The choice of the baseline (reference) category of the response variable isn't crucial. Whichever the category we choose, the model will always imply the same results for all pairs of response categories. The exponential of the individual coefficients can be interpreted as multiplicative effects on the ratios of probabilities π_{il}/π_{i1}. It is evident that the above equation can be inverted to obtain:

$$\pi_{il} = \frac{exp(X_i'\beta_l)}{\sum_{l=1}^{L} exp(X_i'\beta_l)} = \frac{exp(X_i'\beta_l)}{1 + \sum_{l=2}^{L} exp(X_i'\beta_l)} \tag{13}$$

Where we take $\beta_1 = 0$.

In this research context, the following multinomial logistic regression model can be considered, where x, m_1, and m_2 represent dummy variables for the categories of X, M_1, and M_2:

$$P(Y = L|X, M_1, M_2) = \frac{\exp\left(\boldsymbol{\beta}_{l0} + \boldsymbol{x}^\top \boldsymbol{\beta}_{lx} + \boldsymbol{m_1}^\top \boldsymbol{\beta}_{lm_1} + \boldsymbol{m_2}^\top \boldsymbol{\beta}_{lm_2}\right)}{\sum_{l=2}^{L} \exp\left(\boldsymbol{\beta}_{l0} + \boldsymbol{x}^\top \boldsymbol{\beta}_{lx} + \boldsymbol{m_1}^\top \boldsymbol{\beta}_{lm_1} + \boldsymbol{m_2}^\top \boldsymbol{\beta}_{lm_2}\right)} \tag{14}$$

where $x = [x_1, x_2, ..., x_j]^T$, $m_1 = [m_{11}, m_{12}, ..., m_{1k_1}]^T$, $m_2 = [m_{21}, m_{22}, ..., m_{2k_2}]^T$, the values of X, M_1, M_2 and Y are denoted by $j = 1, ..., J$, $k_1 = 1, ..., K_1$, $k_2 = 1, ..., K_2$ and $l = 1, ..., L$. In this paper, $J = L = 7$, $K_1 = 5$, $K_2 = 4$

Interpreting the model based on coefficients can be quite laborious. Therefore, fitted probabilities are particularly valuable as they directly convey the likelihood of each outcome under scenarios defined by the predictor variables. The path analysis begins by estimating conditional distributions, such as $P(Y|X, M_1, M_2)$. This analysis employs a non-saturated model, specifically a multinomial logistic model, which does not include interactions between different variables.

4 Simulation Study

The simulation study was designed to validate the performance of the proposed model, ensuring it aligns with theoretical expectations. The primary goals of this study were to assess the convergence of the model, examine the distribution of the calculated effects, and verify the behavior of the log odds ratios under controlled conditions.

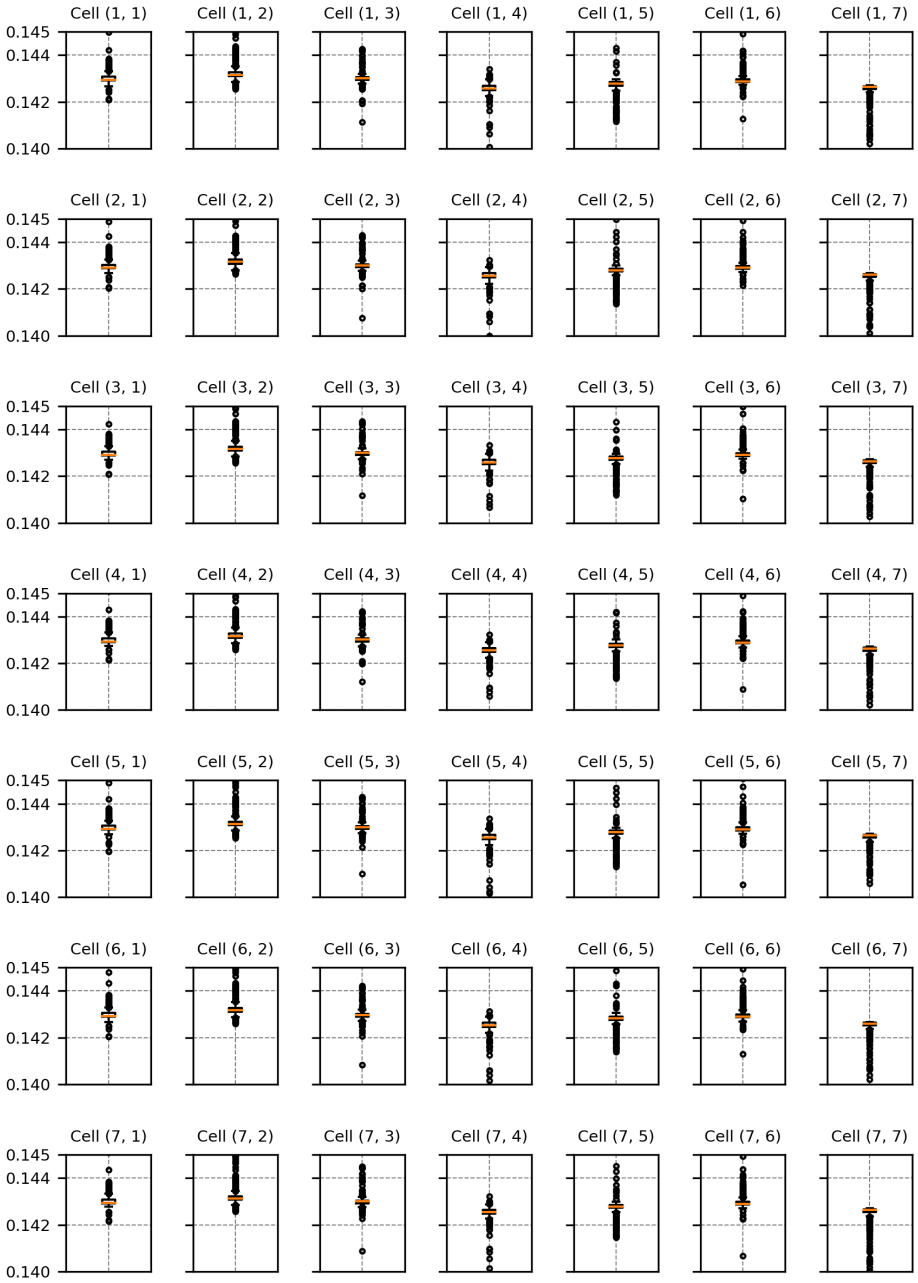
4.1 Data Generation

The simulation employed synthetic datasets generated under controlled conditions. Each variable—X, M_1, M_2, and Y—was assigned values based on a uniform distribution. Specifically, X and Y were categorized into 7 levels, M_1 into 5 levels, and M_2 into 4 levels. This structure mirrors that of the empirical study, ensuring comparability. The use of a uniform distribution implies that each category within a variable had an equal probability of being selected, resulting in a dataset where transitions between categories were uniformly distributed and independent.

4.2 Simulation Analysis

In this process, two main procedures were conducted. First, the analysis involved 1,000 iterations for 5,000 data points, generating four effect tables—total effect, indirect effect 1, indirect effect 2, and indirect effect 3. Each table was structured as a 7x7 matrix. The focus was on examining the distribution and convergence of the values in the 49 cells of these matrices. As shown in Figure 3 (an example for $X \rightarrow M_1 \rightarrow Y$, other paths results are similar), the values across all cells are concentrated around 0.143, which is very close to the theoretical value, equals to $\frac{1}{7} = 0.1428$. This convergence aligns with the theoretical expectation that, under a uniform distribution, the probabilities of transitioning between any two categories should be equal, demonstrating strong stability. Additionally, there are few outliers in the cells, which may be attributed to the randomness in data generation, but they have a minimal overall impact.

Secondly, the log odds ratios for three distinct indirect effects were calculated using equation (6) - equation (8). Given that the values within each cell are similar, the log

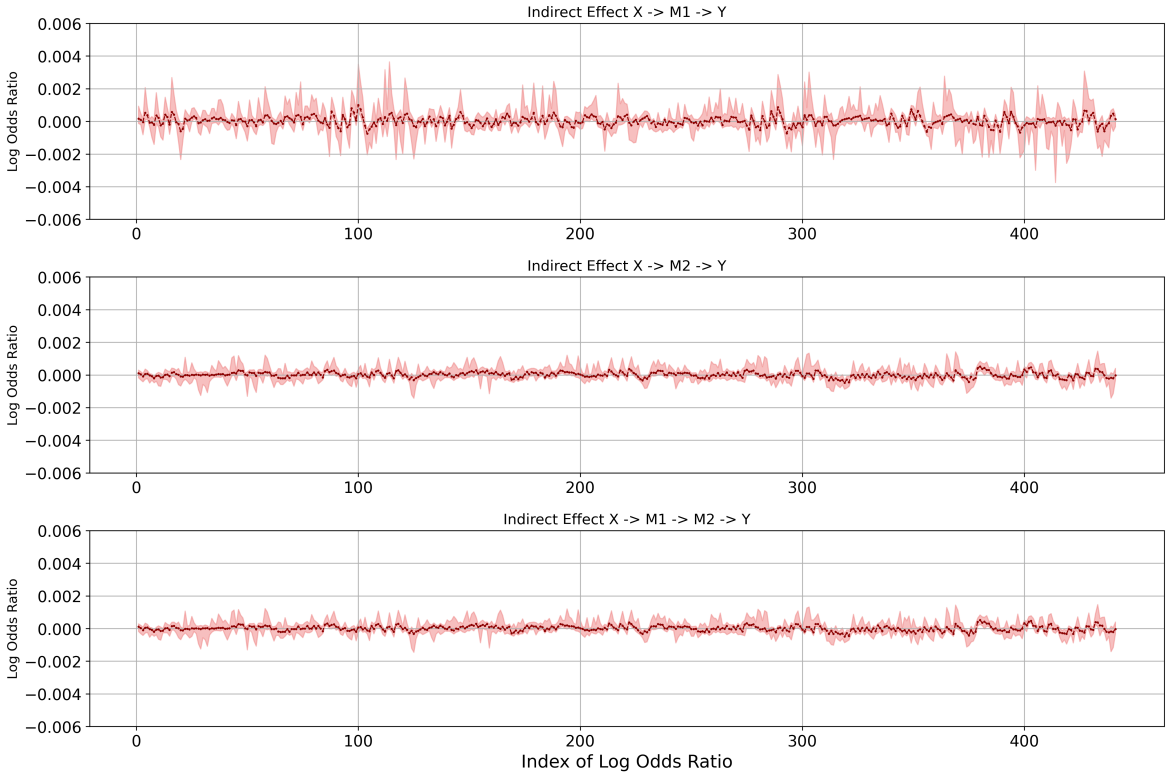


The 49 cells' values in the matrices: The edges of the box represent the first quartile (Q1) and third quartile (Q3), with the orange line in the middle indicating the median. The points outside the whiskers are considered outliers. Most data points are concentrated within the whiskers, around 0.143.

Figure 3: Simulated Transition Probabilities for $X \rightarrow M_1 \rightarrow Y$

odds ratios are theoretically expected to be close to zero, as the log odds ratio is a measure of the strength of association between two categorical variables. Accordingly, 441 log odds ratios $\left(\binom{7}{2} \times \binom{7}{2}\right)$ were calculated for each indirect effect. To further evaluate the precision and variability of these log odds ratios, bootstrap sampling was applied to

the 1,000 generated tables for each effect, resulting in 95% confidence intervals for each of the 441 log odds ratios. These confidence intervals provided a means to assess the stability of the estimated log odds ratios, confirming that they were centered around zero, as theoretically expected under a uniform distribution. This analysis reinforced the robustness of the model and the accuracy of its predictions.



Each path's chart has the index of the Log Odds Ratios on the horizontal axis and the Log Odds Ratio values on the vertical axis. The shaded areas represent the 95% confidence intervals for each Log Odds Ratio. The confidence intervals for almost all Log Odds Ratios fall within the range of ± 0.002.

Figure 4: Log Odds Ratios and 95% Confidence Intervals

Overall, the simulation demonstrated that the model's predictions are consistent with theoretical expectations, showing convergence and stability across multiple simulated datasets.

5 Empirical Analysis

After outlining the methodology and simulation, the empirical analysis was conducted with X representing the class of origin, M_1 representing IQ (intelligence), M_2 representing education, and Y representing the class of destination. The primary objective is to evaluate the extent to which the attainment of M_1 and M_2 mediates the influence of X on Y through various paths across different cohorts, genders, and other subgroups. Each cohort is represented by four tables: one for the total effect and three for the indirect effects. All effects are calculated using the complete data for each combination of variables, without any imputation. This section presents sample tables for the 1970 cohort, with the remaining tables and figures are provided in Appendix A.

5.1 Different Cohort Analysis

This section presents the overall cohort analysis without dividing by gender. It considers both the indirect and total effects as the probabilities of transitioning from one category of X to another category of Y. The discussion will begin with this 7×7 table. The distinction between x and x^* is crucial. The reference distribution P_0 represents the distribution of hypothetical values that do not depend on other variables in the model. This distribution is used to ensure that the calculation of indirect and direct effects is based on a consistent and unbiased baseline. Mathematically, for a given mediator M_1 and outcome Y, the reference distribution P_0 can be expressed as follows: $P_0(M_2 = m_2) = \frac{1}{|M_2|}$, where $|M_2|$ denotes the number of possible levels of the mediator M_2. Similarly, for the treatment variable X, the reference distribution P_0 is defined as: $P_0(X = x) = \frac{1}{|X|}$, where $|X|$ represents the number of possible levels. This ensures that each level of the exposure variable and mediator is equally likely under the reference distribution, providing a neutral baseline for effect estimation. The choice of the reference distribution is not critical, as previous studies, such as Kuha et al. (2021), have shown that varying the reference distribution does not significantly affect the results. The analysis begins with the estimated probabilities presented below. Table 2 illustrates the total effects and indirect effects for the 1970 cohort. The log odds ratios will be computed based on these conditional probabilities.

Table 2: 1970 Cohort 'Total' and 'Indirect' Probabilities

Estimated conditional probabilities of class of origin and class of destination for 1970 cohort. The upper table shows the probabilities $\pi_y(x)$ and the $X \to M_1 \to Y$ indirect effect probabilities, while the lower table displays the $X \to M_2 \to Y$ and $X \to M_1 \to M_2 \to Y$ indirect effect probabilities.

	Destination													
	Total Effect							Indirect Effect$_{X \to M_1 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.39	.29	.15	.06	.02	.06	.03	.23	.26	.15	.11	.06	.11	.08
2	.28	.31	.16	.08	.05	.08	.04	.23	.25	.15	.11	.06	.11	.09
3	.24	.26	.18	.11	.05	.06	.10	.21	.24	.16	.11	.06	.12	.10
4	.17	.22	.15	.18	.06	.12	.10	.20	.24	.16	.12	.06	.12	.10
5	.20	.23	.14	.11	.11	.13	.08	.20	.24	.16	.12	.06	.12	.10
6	.15	.21	.16	.12	.09	.17	.10	.19	.24	.16	.12	.06	.13	.10
7	.12	.20	.16	.11	.07	.15	.19	.19	.23	.17	.12	.06	.13	.10

	Destination													
	Indirect Effect$_{X \to M_2 \to Y}$							Indirect Effect$_{X \to M_1 \to M_2 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.26	.24	.16	.10	.06	.10	.08	.28	.26	.16	.10	.05	.09	.06
2	.23	.24	.17	.11	.06	.11	.08	.24	.26	.16	.11	.06	.10	.07
3	.20	.24	.17	.12	.06	.12	.09	.20	.26	.17	.12	.06	.11	.08
4	.20	.23	.16	.12	.07	.13	.09	.20	.24	.17	.12	.06	.12	.09
5	.19	.23	.17	.12	.07	.13	.09	.19	.24	.17	.12	.07	.12	.09
6	.19	.23	.17	.12	.07	.13	.09	.18	.24	.17	.12	.07	.12	.10
7	.18	.23	.17	.12	.07	.13	.10	.17	.23	.17	.13	.07	.13	.10

As defined by the NS-SEC, Classes 1 and 2 are at the top of the class hierarchy, collectively referred to as the salariat. Classes 6 and 7 are at the bottom, known as the working class. Classes 3, 4, and 5 represent the intermediate classes with 'mixed' employment relationships (the analysis doesn't treat the different classes as ordered variables, but the classification of NS-SEC will help to understand). In 1970, from the perspective of the estimated total probabilities, the probability of mobility from class 1 to class 1 was the highest (0.39), followed by the probability of mobility to class 2 (0.29). The probability of mobility to class 4 and below was significantly lower, indicating a

strong tendency for class 1 to maintain its own class. Overall, the mobility between class 1 and class 2 (salariat) was relatively high, demonstrating a strong class retention, while the mobility of classes 6 and 7 was more dispersed. The 1946 and 1958 cohorts (results see Appendix A) also show that the internal mobility rates within classes 1 to 3 are higher, indicating that individuals originally in classes 1 to 3 are more likely to remain within this class range when determining their class of destination. There are some differences between the overall cohort analysis and the subsequent gender analysis. In the overall cohort analysis, the probability of individuals in class 1 moving to class 2 was higher than maintenance (the value of cell $(1, 1)$ is smaller than the value of cell $(1, 2)$), particularly evident in the 1946 and 1958 cohorts, but less pronounced in the 1970 cohort. This higher mobility from class 1 to class 2 in the overall cohort is largely driven by the patterns seen in women. When analyzing within the same cohort by gender, it was observed that men exhibited stronger class 1 retention across various effects, whereas women had a higher probability of moving from class 1 to class 2.

The log odds ratio is a measure used to determine the strength of association or non-independence between two categorical variables. With the log odds ratio defined, a total of 441 log ORs are utilized to summarize various aspects of social mobility. In the context of social mobility, an OR equal to 1 signifies perfect mobility, where movement between classes is entirely open and unrestricted. ORs that deviate further from 1 indicate greater inequality in relative mobility opportunities, or lower 'fluidity' within the class structure, suggesting a barrier to mobility. Of particular interest is the subset of log ORs known as diagonal log odds ratios. The Figure 5 below illustrates the concept of diagonal log odds ratios, which compare identical origin and destination classes. For instance, among the 49 cells in the table, the diagonal log odds ratio compares class 1 with class 7. This process involves extracting the four corners of the indirect effect table to form a 2×2 table, and then applying the formula to measure the differences.

For the 1970 cohort, the total log odds ratio (1 vs 7) is the highest, at 3.02, indicating significant barriers to mobility between these two extreme categories and reflecting the phenomenon of social class rigidity. The log odds ratios between Class 1 and

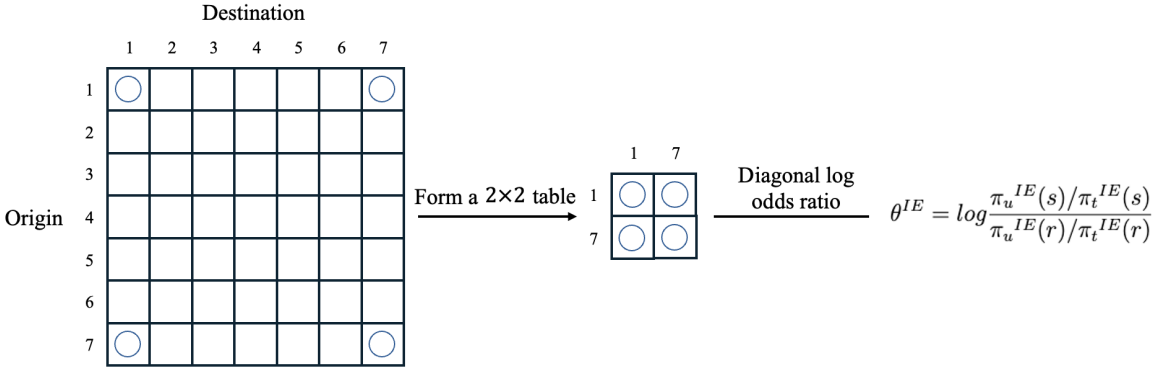


Figure 5: Diagram of Diagonal Log Odds Ratio '1 vs 7'

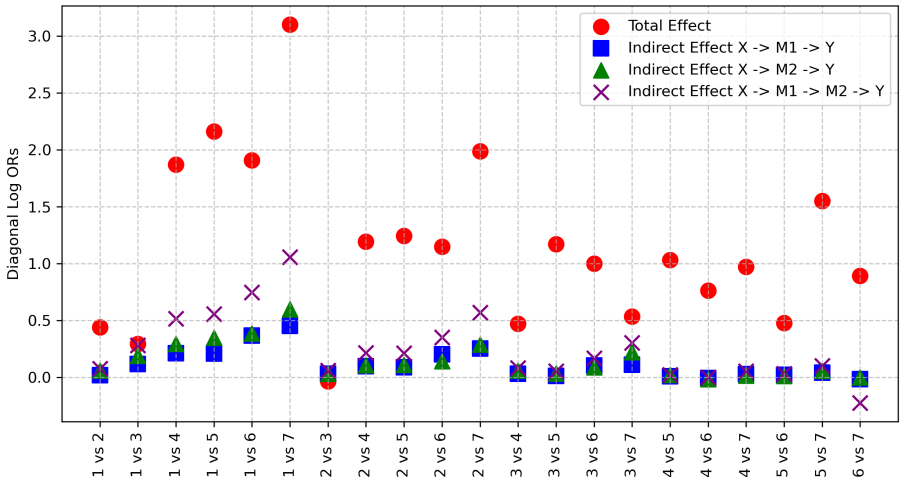
other categories are also relatively high, particularly with Class 5 and Class 6, at 2.37 and 1.99 respectively, suggesting limited mobility between high-salary occupations and these categories. In contrast, the log odds ratio between Class 2 and Class 3 is relatively low, at 0.29, indicating greater mobility within these middle-income occupations.

Considering the indirect effect between Class 1 and Class 7, the indirect log odds ratio for the path $X \to M_1 \to Y$ is 0.41, for the path $X \to M_2 \to Y$ is 0.59, and for the path $X \to M_1 \to M_2 \to Y$ is 1.01. This suggests that the IQ variable alone accounts for only 13.7% of the total effect (in the 1946 cohort, the IQ variable alone contributes 11.2%, and in 1958, it accounts for 18.4%). These figures indicate that, despite differences across cohorts, the IQ variable alone is insufficient to strongly explain the total association. The education variable alone contributes 19.5%, which is slightly higher than the contribution from the IQ variable alone. The combined path involving both IQ and education variables accounts for 33.4% of the total effect. While this combined effect is greater than the contributions from the individual variables, it still does not fully explain the total association (or lack of mobility) between these classes.

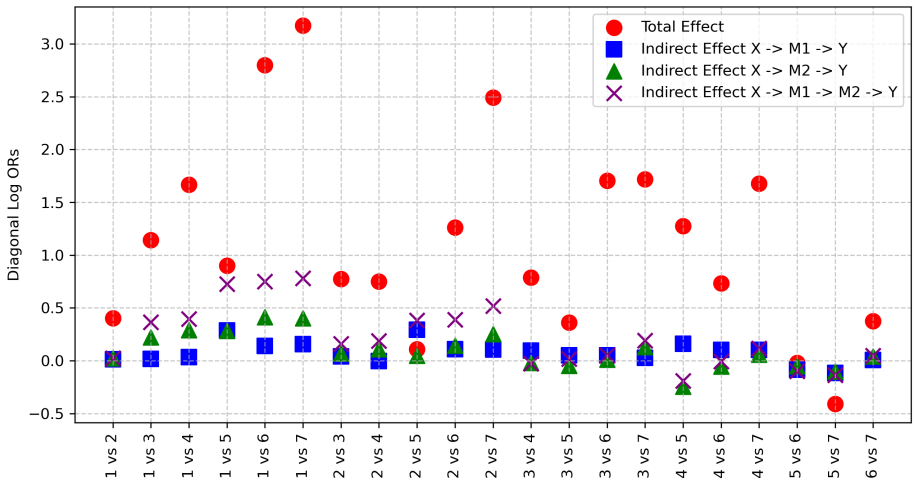
5.2 Different Gender Analysis

This section examines gender differences across the three cohorts by estimating and visualizing the diagonal log odds ratios for different paths, separately for men and women. Figure 6a and Figure 6b display the estimated total and indirect log odds ratios (across three paths) for men and women in the 1970 cohort, with a focus on the

21 diagonal log odds ratios. The results clearly show that as the hierarchical distance increases, the total association becomes more pronounced. This pattern is also observed in the indirect associations.



(a) Men



(b) Women

21 estimated diagonal log odds ratios between class of origin and class of destination, separated by gender across the 1970 cohort. Three separate indirect effect pathways as well as the total effect pathway are shown. For instance, in the upper plot, the diagonal log ORs involving classes 1 and 7 for men in the 1970 cohort are illustrated in the "1 vs 7" section. The total log OR is 3.10 marked by a red circle, while the three indirect effects are represented by other shapes, with values of 0.45, 0.59, and 1.06.

Figure 6: Log ORs for Different Effects in 1970 Cohort, by Gender

Firstly, across the three cohorts, as well as within both male and female subgroups, the diagonal log odds ratio is consistently highest for the $X \to M_1 \to M_2 \to Y$ path, followed by the $X \to M_2 \to Y$ path, and lowest for the $X \to M_1 \to Y$ path. Specifically, in the 1970 cohort, the "1 vs 7" comparison shows a high log odds ratio for both men (3.10) and women (3.17), indicating a strong barrier to mobility. For the "1 vs 7" comparison, the proportions of the total effect's log OR attributed to the three indirect paths are 14.5%, 19.03%, and 33.98% for men, and 4.73%, 12.62%, and 24.61% for women. This suggests that IQ has a stronger explanatory power for men, aligning with the original data showing higher IQ levels among men in the 1970 cohort (as discussed in Section 2).

In the 1946 cohort (see Figure 8), the total effect diagonal log odds ratios for men in the "1 vs 2" (0.17) and "2 vs 3" (0.21) comparisons are close to zero, indicating greater mobility. For women in the 1946 cohort, the total ORs are above 10 for the "2 vs 5" and "4 vs 7" comparisons, reflecting significant social mobility barriers. Additionally, the explanatory power of the three indirect paths is weak for women when transitioning from class 3 to classes 4 through 7. For example, in the "3 vs 4" comparison, the total log OR is 3.33, while the log ORs for the three indirect paths are only 0.03, 0.13, and 0.11, respectively. However, the indirect effects explain the transitions from class 1 and class 2 to lower classes more effectively, suggesting that social mobility for women in the 1946 cohort, starting from class 3, is likely influenced by factors beyond IQ and education. In the 1958 cohort, men exhibit higher mobility barriers in the "1 vs 6" and "1 vs 7" comparisons, with total log ORs of 3.47 and 3.46, respectively. The data indicates that the factors influencing male transitions from class 4 to lower classes are not well explained by IQ and education variables. In contrast, the patterns for women in the 1958 cohort are less consistent. Although the $X \to M_1 \to M_2 \to Y$ path still has the highest explanatory power, in later class transitions, IQ alone often matches or even surpasses education in explanatory power, highlighting the increasing importance of IQ for female mobility in these classes.

Building on these findings, the analysis further focuses on examining the characteristics

of the estimated ratios between each distinct indirect effect and the total effect across different cohorts and gender subsets. Specifically, attention is given to the ratios

$$\hat{\text{Ratio}}_1 = \hat{\theta}^{IE_1}_{rs,tu} / \hat{\theta}^{TE}_{rs,tu} \quad (\text{for } X \to M_1 \to Y) \tag{15}$$

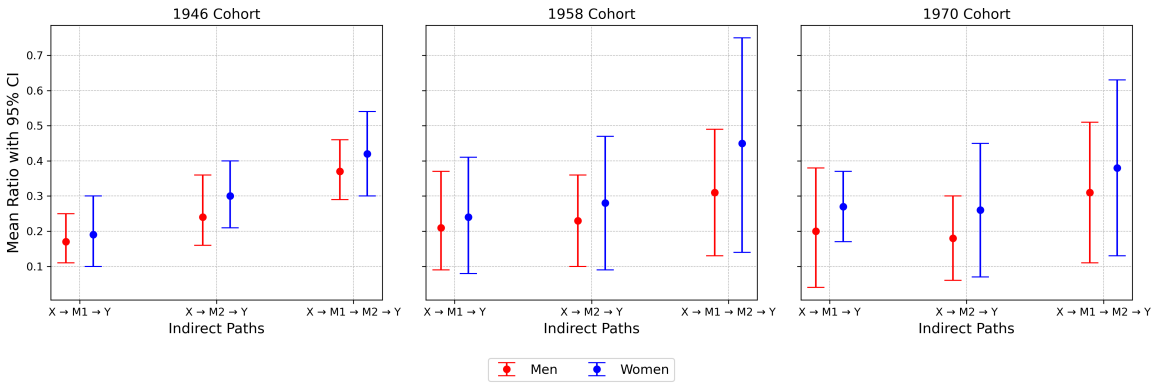
$$\hat{\text{Ratio}}_2 = \hat{\theta}^{IE_2}_{rs,tu} / \hat{\theta}^{TE}_{rs,tu} \quad (\text{for } X \to M_2 \to Y) \tag{16}$$

$$\hat{\text{Ratio}}_3 = \hat{\theta}^{IE_3}_{rs,tu} / \hat{\theta}^{TE}_{rs,tu} \quad (\text{for } X \to M_1 \to M_2 \to Y) \tag{17}$$

which represent the proportion of each indirect effect relative to the total log odds ratio for the origin and destination classes.

To ensure the robustness and reliability of the findings, 95% confidence intervals for the estimated ratios were calculated. In certain instances, the estimated total log odds ratios were found to be extremely small, potentially obscuring the overall patterns. This issue may stem from high sampling variation or situations where the total population association is close to zero. To address this concern, the analysis focuses on cases where the total association is not negligible, specifically when the Wald test p-value of the total log odds ratio is less than 0.1, indicating statistical significance. Furthermore, cases with negative total log odds ratios are excluded to emphasize positive associations. This approach enables a more focused analysis on meaningful associations, thereby enhancing the understanding of how the indirect effects contribute to the total effect. The bootstrap resampling method of variance estimation can be used, assuming the estimated probabilities are asymptotically normally distributed, is employed to calculate the standard errors and estimate the confidence intervals of these ratios. The detailed steps for this method are outlined in Appendix B.

Based on Figure 7, among the 882 ratios analyzed (441 for men and 441 for women), the overall pattern shows that in the paths $X \to M_2 \to Y$ and $X \to M_1 \to M_2 \to Y$, the mean ratio for women is significantly higher than that for men. In contrast, in the path $X \to M_1 \to Y$, the mean ratios for men and women are relatively similar. Additionally, the 95% confidence intervals for women are generally larger than those for men (more evident in the 1958 and 1970 cohorts), except for the 1946 cohort, where the confidence interval for women is slightly smaller. This may be due to the larger number of missing values for men in 1946, leading to errors (as multiple imputation was



Summary statistics of the estimated ratios between three distinct indirect and total log odds ratios for men and women across the three birth cohorts.

Figure 7: Confidence Interval of Estimated Ratios

not employed in this empirical analysis). From Section 2, it is noted that the average IQ for the 1970 cohort was 0.09 (after standardization), significantly higher than in 1958 (-0.01) and 1946 (-0.02). However, this increase in IQ does not manifest in the mean ratio for the indirect effect driven by the IQ variable alone, as the mean ratio remains similar between the 1970 cohort (0.20 for men and 0.27 for women) and the 1958 cohort (0.21 for men and 0.24 for women). This indicates that the IQ variable alone does not significantly contribute to overall social mobility, but it has a substantial impact on long-distance mobility, such as the "1 vs 7" comparison, as demonstrated in the previous gender-specific analysis. When focusing on women, the contribution of IQ to their social mobility exhibits an overall increasing trend.

Table 3 presents the summary statistics for each combination of cohort and gender. The average ratio for men ranges from 0.19 to 0.32, while for women it ranges from 0.22 to 0.40. The data shows that, across all cohorts and different genders, the mean ratio for women is slightly higher than for men (except for the IQ variable alone), and the confidence intervals are wider. This is consistent with analysis above. Among the different cohorts, there is no clear trend in the $X \to M_1 \to Y$ and $X \to M_2 \to Y$ paths, while the $X \to M_1 \to M_2 \to Y$ path shows a slight downward trend, with a decrease of only 0.04 between 1946 and 1970 cohort.

Table 3: Summary Statistics of Estimated Ratios

Summary statistics of the estimated ratios categorized by cohort or gender. The upper part of the table is divided by different cohorts according to the year, without distinguishing gender, and shows the significant ratios for each distinct indirect effect. The lower part of the table aggregates all the data without considering the year and instead focuses solely on gender differences.

Separately by cohort:									
	1946 cohort			**1958 cohort**			**1970 cohort**		
	IE_1	IE_2	IE_3	IE_1	IE_2	IE_3	IE_1	IE_2	IE_3
No.	88	100	115	88	100	115	88	100	115
Mean	.20	.24	.39	.22	.27	.37	.21	.20	.35
C.I.	(.09-.33)	(.11-.37)	(.20-.60)	(.09-.35)	(.11-.45)	(.13-.62)	(.06-.35)	(.06-.34)	(.13-.56)

Separately by gender:						
	Men			**Women**		
	IE_1	IE_2	IE_3	IE_1	IE_2	IE_3
No.	168	169	179	168	169	179
Mean	.20	.19	.32	.22	.27	.40
C.I.	(.09-.30)	(.08-.30)	(.15-.50)	(.06-.34)	(.07-.48)	(.12-.60)

"No". means number of Log ORs for which $p < 0.10$ *for the estimated association.*

"C.I". means the range between 5th and 95th percentiles of the estimated ratios.

IE_1 *represents* $X \to M_1 \to Y$, IE_2 *represents* $X \to M_2 \to Y$, IE_3 *represents* $X_1 \to M_2 \to Y$

5.3 Class Transition Analysis

This section explores the variation in ratios across different types of class transitions and fits a linear regression model, consistent with the approach of Kuha et al. (2021). The response variable is the estimated ratio calculated earlier, with eight explanatory variables as detailed below.

The first explanatory variable (X_1) is the total log odds ratio itself. The second one is "White-collar to/from Other" (X_2). This is a binary (0/1) indicator variable used to mark whether social mobility has crossed the boundary between the "white-collar" classes (classes 1-3) and other classes (classes 4-7). Specifically, if $r, t \leq 3$ and $s, u > 3$, this variable is set to 1, indicating that social mobility has occurred across the boundary between the "white-collar" classes and other classes. Otherwise, the variable is

set to 0. The third and fourth explanatory variables measure "inheritance effects". "Inheritance effects" refer to influences that reinforce social immobility and thereby constrain mobility, stemming from the particular appeal that positions within one's own class of origin hold for individuals (Bukodi et al. (2017)). Two variables are used to measure the effects. The third variable (X_3) captures the maintenance of class 1, where $r = t = 1$, and is a binary (0/1) variable. The fourth variable (X_4) captures "inheritance effects" for all other classes, specifically referring to cases where the origin and destination classes are the same but not equal to class 1. A value of 0 indicates that no inheritance effect has occurred (i.e., the origin and destination classes are not the same). A value of 1 indicates that an inheritance effect has occurred once (i.e., the origin and destination classes are the same), and a value of 2 indicates that inheritance effects have occurred in two different combinations.

Finally, the last four explanatory variables ($X_5 - X_8$) are used to measure "hierarchy effects". "Hierarchy effects" zrelate to the availability of family economic, cultural, and social resources, which may act as barriers to reaching the destination class. Although there are originally seven classes, here classes 3, 4, and 5 are treated as one single category, effectively reducing the total to five hierarchical positions. This leads to four possible hierarchical distances, corresponding to the four explanatory variables. The hierarchical positions are defined as $h_c = \{1, 2, 3, 3, 3, 4, 5\}$, where classes 3, 4, and 5 share the same level. The hierarchical distance for a given transition is calculated as:

$$X_{\delta+4} = |I_\delta(r,t) - I_\delta(r,u) - I_\delta(s,t) + I_\delta(s,u)| \tag{18}$$

Here, $\delta = 1, 2, 3, 4$ represents the level of hierarchical distance. The indicator function $I_\delta(o,d)$ takes the value of 1 if the absolute difference between the hierarchical positions of the origin class h_o and the destination class h_d is greater than or equal to δ, and 0 otherwise:

$$I_\delta(o,d) = \begin{cases} 1 & \text{if } |h_o - h_d| \geq \delta \\ 0 & \text{otherwise} \end{cases}$$

The variable x_δ can take values of 0, 1, or 2.

Table 4: Estimated Coefficients for $X \to M_1 \to M_2 \to Y$

*Estimated coefficients of linear regression models for the estimated ratios between indirect effects and total effects in the $X \to M_1 \to M_2 \to Y$ path, fitted separately for men and women in each birth cohort. The model is based on 1000 bootstrap resampling methods. Significance codes are as follows: "***" indicates $p \approx 0$, "**" indicates $p < 0.001$, "*" indicates $p < 0.01$, and "·" indicates $p < 0.05$.*

Explanatory variable	**1946 cohort**		**1958 cohort**		**1970 cohort**	
	Men	Women	Men	Women	Men	Women
(constant)	1.349***	0.497***	0.357***	1.199***	0.309***	0.453***
X_1	-0.581***	-0.030***	-0.209***	-0.056*	-0.225***	-0.143***
X_2	0.068**	0.010	0.115**	0.031	0.187***	-0.027
Inheritance effect:						
X_3	0.375 ·	-0.196 ·	0.015	0.670*	0.178***	0.143**
X_4	-0.435*	0.045	-0.096 ·	-0.595*	-0.138***	-0.061**
Hierarchical effect:						
X_5	-0.030	0.146**	0.107**	-0.290 ·	-0.063***	-0.104**
X_6	-0.030	0.015	0.028	-0.227*	0.058***	0.029
X_7	0.239*	0.024	0.096**	-0.354*	0.068***	0.056 ·
X_8	0.199	-0.134 ·	0.092*	0.436 ·	0.103***	0.043 ·
R^2	0.59	0.41	0.49	0.47	0.75	0.67

Explanatory Variables X_1-X_8 are defined above.

Table 4 shows that based on the R^2 values, the $X \to M_1 \to M_2 \to Y$ path demonstrates the highest explanatory power for the estimated ratios among the variables. Across the three cohorts, the coefficient of X_1 (total log OR) consistently shows negative and is highly significant, indicating that as the total log OR increases, the estimated ratios decrease. This finding suggests why the indirect effect's contribution to the total effect is limited, with the maximum contribution not exceeding 50%. The coefficient for X_2 (White-collar to/from Other) is positive, means that when an individual moves from a white-collar class to another class, the indirect association becomes more pronounced, with a stronger effect observed in men than in women. Furthermore, in the context of inheritance effects, men are more likely to remain in class 1 (X_3 is positive), whereas when more individuals are involved in transitions within classes 2 to 7, the indirect association tends to decrease in relation to the total association. The hierarchical effect is generally positive; however, an exception is found among women in the 1958

cohort, where the coefficient only turns positive when $\delta = 4$.

Table 5: Estimated Coefficients for $X \to M_1 \to Y$

Explanatory variable	**1946 cohort**		**1958 cohort**		**1970 cohort**	
	Men	Women	Men	Women	Men	Women
(constant)	1.266***	0.296*	0.898***	0.683***	0.177***	0.082***
X_1	-0.501***	0.001	-0.484***	0.025*	-0.112***	-0.035*
X_2	-0.421 ·	-0.260*	-0.191	0.120	-0.075***	-0.376*
Inheritance effect:						
X_3	-0.038	-0.075	0.345 ·	0.522*	0.085***	0.001
X_4	0.103	0.255	-0.214	-0.291 ·	-0.051***	0.012
Hierarchical effect:						
X_5	0.044	0.010	-0.073	-0.347*	-0.050***	-0.018
X_6	-0.036 ·	0.184 ·	0.058	-0.134	0.028***	-0.352*
X_7	0.073	0.304*	0.241*	-0.365**	0.033***	0.014 ·
X_8	0.454*	0.121	0.069*	0.517**	0.041***	0.023*
R^2	0.42	0.33	0.40	0.52	0.71	0.66

Significant codes are same as above.

Table 6: Estimated Coefficients for $X \to M_2 \to Y$

Explanatory variable	**1946 cohort**		**1958 cohort**		**1970 cohort**	
	Men	Women	Men	Women	Men	Women
(constant)	0.664***	0.349***	0.412***	1.094***	0.198***	0.218***
X_1	-0.350***	-0.016***	-0.280***	-0.749***	-0.126***	-0.049**
X_2	0.009	-0.008	0.078	0.245	0.078***	-0.050
Inheritance effect:						
X_3	0.115	-0.146·	-0.013	0.209***	0.092***	-0.043***
X_4	-0.171*	0.070	-0.064 ·	-0.399 ·	-0.091***	-0.047*
Hierarchical effect:						
X_5	0.061	0.045	0.089*	-0.744	-0.016***	0.046***
X_6	0.010	0.038	0.060	0.048**	0.029***	0.016
X_7	0.216**	0.045*	0.062 ·	0.065	0.033***	0.034 ·
X_8	0.099 ·	0.037	0.141*	0.074*	0.062***	0.004
R^2	0.52	0.50	0.42	0.45	0.73	0.56

Significant codes are same as above.

For the $X \rightarrow M_1 \rightarrow Y$ and $X \rightarrow M_2 \rightarrow Y$ paths, X_1 (total log OR) also shows consistently negative and highly significant coefficients. The overall characteristics of $X \rightarrow M_2 \rightarrow Y$ are similar to those of $X \rightarrow M_1 \rightarrow M_2 \rightarrow Y$. However, in the $X \rightarrow M_1 \rightarrow Y$ path, the coefficients for X_2 are more often negative, suggesting that when more individuals experience mobility transitions between white-collar and other classes, the estimated ratios are smaller. This may be because the indirect association is close to zero among these class transitions. However, the X_2 coefficient for women in the 1958 cohort is positive, which indirectly indicates the greater importance of IQ for women in 1958, consistent with the gender analysis.

The main findings of the empirical study are as follows. First, overall, the contribution of the three indirect paths to the total effect is less than half, indicating that factors other than IQ and education also play a role. Second, the importance of the three paths follows the order: $X \rightarrow M_1 \rightarrow M_2 \rightarrow Y$, $X \rightarrow M_2 \rightarrow Y$, $X \rightarrow M_1 \rightarrow Y$. Although the explanatory power of IQ alone is relatively weak, in certain cases, the IQ variable plays a significant role, particularly in long hierarchical distance transitions (such as "1 vs 7"), where the explanatory power of the IQ path increases. Additionally, the contribution of IQ to women's social mobility shows an increasing trend, especially among women in the 1946 and 1958 cohorts. Third, when comparing by gender, the mediating variables have a stronger impact on the attainment of class of origin and class of destination for women than for men. Fourth, in different class transitions, men show stronger retention in class 1 compared to women. The mobility between the salariat classes (class 1 and class 2) is relatively strong, whereas the explanatory power of the mediating variables in transitions involving classes 4 to 7 is limited.

6 Conclusion

This paper analyzed the role of pathways with IQ and education in British social mobility. The study found that none of the indirect paths accounted for more than 50% of the total effect, and no significant trends were observed across different cohorts and genders, although the mediating variables had a slightly stronger impact on women than on men. For the newly introduced IQ variable in this study, IQ alone had the least explanatory power (with the smallest estimated ratio). However, in the third pathway, where IQ influences education and then the class of destination, the estimated ratio was higher than that of education alone, suggesting that IQ may have a facilitative effect on educational attainment. Additionally, IQ played a more important role in long hierarchical distance social mobility. Methodologically, the study treated IQ and education as two distinct categorical variables and did not consider the interaction between the two mediating variables (no bidirectional arrows). Conditional probabilities were used to quantify various effects, with focus on association rather than causal effect.

There are several noteworthy future directions for the present work. First, regarding categorical and continuous data: In this study, the IQ variable, originally a continuous variable, was divided into five categories. Thus, a first extension would involve treating IQ as a continuous variable, specifically transforming the summation of the M_1 variable into integration, with preliminary formulas provided in Appendix C. Second, in terms of data processing, we only used complete data, where missing values were directly excluded. Future research should consider the use of multiple imputation methods to further ensure data robustness. Third, additional mediators need to be developed. The explanatory power of the indirect paths in this study is insufficient, suggesting the potential existence of other mediators. Fourth, the dataset used is limited to the period before 1970. Recent studies have shown a trend towards extending the time coverage, such as Paterson (2021), who found that the average intelligence in the highest-status classes may have moved closer to the mean nowadays. Expanding the time range of the dataset is also a potential direction for future research.

7 Appendix

7.1 Appendix A: Results of different effects

The probabilities table is calculated using nested for loops in R. The following are the results for the 1946 cohort and the 1958 cohort.

1946 cohort results:

Table 7: 1946 Cohort 'Total' and 'Indirect' Probabilities

	Destination													
	Total Effect							Indirect Effect$_{X \to M_1 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.17	.53	.12	.11	.02	.03	.02	.10	.33	.22	.10	.06	.10	.09
2	.16	.41	.19	.07	.05	.06	.05	.10	.32	.22	.10	.07	.09	.10
3	.17	.39	.22	.04	.05	.06	.07	.10	.32	.21	.10	.07	.10	.10
4	.10	.34	.13	.21	.06	.08	.08	.10	.30	.20	.11	.08	.10	.11
5	.09	.19	.22	.08	.15	.10	.17	.09	.30	.20	.11	.09	.10	.11
6	.07	.25	.19	.09	.09	.16	.15	.09	.30	.19	.11	.09	.10	.12
7	.06	.19	.20	.07	.10	.15	.23	.09	.30	.19	.11	.09	.10	.12

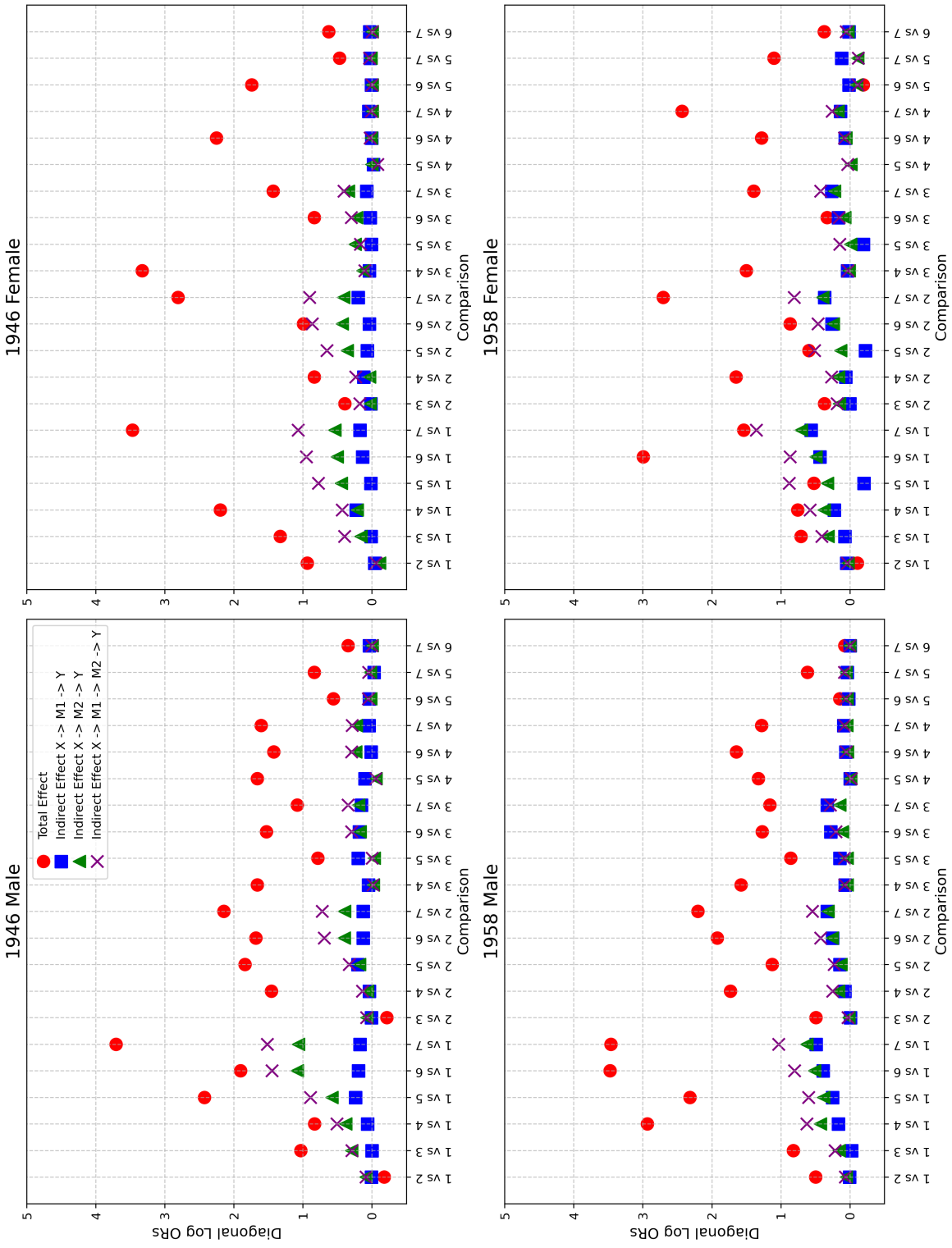
	Destination													
	Indirect Effect$_{X \to M_2 \to Y}$							Indirect Effect$_{X \to M_1 \to M_2 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.15	.35	.15	.10	.08	.08	.09	.17	.39	.14	.09	.08	.07	.06
2	.12	.33	.18	.10	.08	.09	.10	.14	.36	.17	.10	.08	.07	.08
3	.11	.33	.19	.10	.08	.09	.10	.12	.35	.18	.10	.08	.08	.09
4	.10	.31	.20	.11	.08	.09	.11	.10	.32	.21	.10	.08	.09	.10
5	.08	.27	.21	.11	.08	.12	.13	.07	.27	.22	.11	.08	.12	.13
6	.07	.27	.22	.11	.08	.12	.13	.07	.26	.22	.12	.08	.12	.13
7	.07	.27	.21	.11	.08	.13	.13	.07	.26	.21	.12	.08	.12	.14

1958 cohort results:

Table 8: 1958 Cohort 'Total' and 'Indirect' Probabilities

	Destination													
	Total Effect							Indirect Effect$_{X \to M_1 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.33	.32	.16	.07	.04	.03	.05	.16	.25	.19	.13	.07	.10	.10
2	.23	.34	.17	.09	.06	.07	.04	.15	.25	.18	.14	.07	.11	.10
3	.18	.26	.19	.13	.06	.09	.09	.14	.24	.18	.14	.08	.11	.11
4	.11	.20	.10	.29	.06	.11	.13	.14	.24	.16	.14	.08	.12	.12
5	.10	.20	.17	.13	.11	.15	.14	.13	.23	.16	.15	.08	.12	.13
6	.07	.18	.16	.10	.11	.18	.20	.12	.22	.15	.15	.09	.13	.14
7	.08	.17	.13	.11	.08	.19	.24	.12	.22	.15	.16	.09	.13	.13

	Destination													
	Indirect Effect$_{X \to M_2 \to Y}$							Indirect Effect$_{X \to M_1 \to M_2 \to Y}$						
Origin	1	2	3	4	5	6	7	1	2	3	4	5	6	7
1	.17	.26	.15	.13	.07	.11	.11	.20	.29	.15	.12	.07	.09	.08
2	.16	.25	.15	.13	.08	.11	.12	.17	.27	.16	.13	.07	.10	.10
3	.14	.24	.15	.14	.08	.12	.13	.15	.25	.16	.14	.08	.11	.11
4	.13	.23	.16	.14	.08	.12	.14	.13	.23	.16	.15	.08	.12	.13
5	.13	.23	.16	.14	.08	.12	.14	.13	.23	.16	.14	.08	.12	.14
6	.12	.22	.16	.14	.08	.13	.15	.12	.22	.16	.15	.08	.13	.14
7	.12	.22	.16	.14	.08	.13	.15	.11	.21	.16	.15	.08	.14	.15



21 estimated diagonal log odds ratio separated by gender in 1946 cohort and 1958 cohort.

Figure 8: Log ORs for Different Effects in 1946 and 1958 Cohorts, by Gender

7.2 Appendix B: Standard errors for the ratios

The standard errors of the estimated coefficient ratios were assessed using bootstrap resampling methods, assuming that the estimated probabilities are asymptotically normally distributed. The detailed steps are outlined as follows:

- **Data resampling**: The dataset was resampled with replacement to generate new datasets. Each resampled dataset maintains the same size as the original.
- **Generating effect tables and Log ORs**: For each resampled dataset, four contingency tables were recalculated to represent the total effect, indirect effect 1, indirect effect 2, and indirect effect 3. And calculate 441 log odds ratio with equation (9).
- **Computing ratios**: For each combination within the resampled dataset, the ratios of the indirect effects to the total effect were calculated.
- **Repeating the process**: The aforementioned steps were repeated 1000 times to generate a distribution of ratio estimates.
- **Calculating standard errors**: Based on all bootstrap samples, the standard error (SE) for each ratio was determined using the formula:

$$SE = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\hat{Ratio}_i - \bar{\hat{Ratio}}\right)^2} \tag{19}$$

  where $\hat{Ratio}_i$ denotes the ith bootstrap sample's ratio, and $\bar{\hat{Ratio}}$ is the mean ratio across all samples.

7.3 Appendix C: Estimation for continuous situation

This section outlines an approach to treat the IQ variable as a continuous variable, as initially mentioned in Section 2, while other variables remain categorical. The associations of interest will still be expressed as log odds ratios. The main modification to the effect equations involves replacing summations with integrals to account for the continuous nature of IQ, which necessitates evaluating conditional probability densities

over a continuous range.

First, for the pathway $X \to M_1 \to Y$, the equation for the indirect effect is:

$$\pi_y^{IE}(x) = \sum_{x^*} \sum_{m_2} \int P(Y = y \mid X = x^*, M_2 = m_2, M_1 = m_1) \\ P_0(M_2 = m_2) P(M_1 = m_1 \mid X = x)\, dM_1 \cdot P_0(X = x^*) \tag{20}$$

Second, for the pathway $X \to M_2 \to Y$, the indirect effect is given by:

$$\pi_y^{IE}(x) = \sum_{x^*} \sum_{m_2} \int P(Y = y \mid X = x^*, M_2 = m_2, M_1 = m_1) \\ P(M_2 = m_2 \mid X = x, M_1 = m_1) P_0(M_1 = m_1)\, dM_1 \cdot P_0(X = x^*) \tag{21}$$

Third, for the pathway $X \to M_1 \to M_2 \to Y$, the equation is modified to:

$$\pi_y^{IE}(x) = \sum_{x^*} \sum_{m_2} P(Y = y \mid X = x^*, M_2 = m_2, M_1 = m_1^*) \\ P(M_2 = m_2 \mid X = x) P_0(M_1 = m_1^*) P_0(X = x^*), \tag{22}$$

where $P(M_2 = m_2 \mid X = x)$ is derived as:

$$P(M_2 = m_2 \mid X = x) = \int P(M_2 = m_2 \mid M_1 = m_1, X = x) \\ \times P(M_1 = m_1 \mid X = x)\, dM_1 \tag{23}$$

Finally, the equation for the direct effect becomes:

$$\pi_y^{DE}(x) = \sum_{x^*} \sum_{m_2} \int P(Y = y \mid X = x^*, M_2 = m_2, M_1 = m_1) \\ P_0(M_2 = m_2) P_0(M_1 = m_1) P_{x^*}(X = x^*)\, dM_1 \tag{24}$$

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